The $k = 4$ index calculus end to end: the D_4 baseline and the V_4 line

Goal G2, `RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` §9.5–§9.14 · 2026-10-17/18 · experiments E19–E24

1. Result

On curves over $\mathbb{F}_{p^4}$ with full rational 2-torsion, a base line stable under all three translations and S_5 written in the invariants of $V_4^{4_0} \rtimes S_4$ (four unknowns of weights $(2, 2, 1, 1)$, quotient 64, Macaulay degree 11) cut one decomposition from $1.41 \cdot 10^9$ to $1.66 \cdot 10^6$ $\mathbb{F}_p$ multiplications. Measured end to end (E24, 12 curves, $p = 101$–521), **the $k = 4$ index calculus is 35–240 × cheaper than the D_4 baseline (E19, 10 curves) at every matched size, and the modelled handover with rho moves from $2^{116.7}$ to $2^{90.1}$.** Every logarithm was recovered and verified; the exponent is unchanged ($S/$ rho $\propto n^{-1/4}$, fitted -0.25): **engineering, a constant.**

Distance remaining. Every row is still $1.3 \cdot 10^4$–$1.7 \cdot 10^5$ × slower than the matched negation rho. The handover is a model extrapolation of measured constants with the theory exponents, not a measured crossing. **Scope:** full rational 2-torsion with rational halves of two T's, $h = 4$, $n \leq 2^{34.1}$.

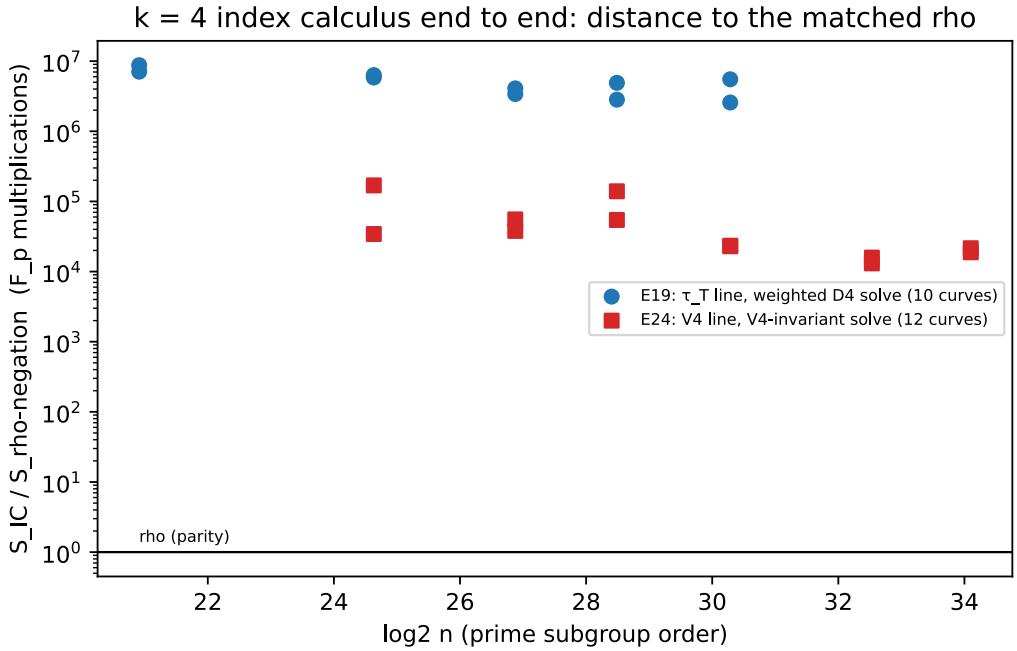


Figure 1: $S/$ rho per curve on both pipelines (E19: 10 curves; E24: 12 curves); 16 negation walks per curve; runs `experiments/27_quartic_pipeline_d4`, `experiments/29_quartic_pipeline_v4`.

2. What ran

Each experiment was pre-registered in the note before its code. E19: the τ_T-folded line and the weighted D_4 solve (degree 26). E24: E19's protocol on the V_4 line and solve (degree 11). Both: uniform targets $R = aG + bQ$; one verified decomposition per target as a folded row; Wiedemann until $[d]G = Q$ verifies; every phase charged (precompute, base, generation, Weil restriction, solves, lift-back, group checks, linear algebra at 16 per mod-n multiplication); 16 negation and 16 unfolded walks on the same curve; unpriced native work at 0, 0.1 and 1 GAE. Host: one shared 4-vCPU container, L0; operation counts only.

p	E19 $S/$ rho	E24 $S/$ rho	ratio (by seed)	E19 C	E24 C	E24 targets
101	$6.33, 5.85 \cdot 10^6$	$1.69 \cdot 10^5, 3.43 \cdot 10^4$	37, 171	$1.39 \cdot 10^9$	$1.62 \cdot 10^6$	77,960 / 15,387
149	$4.09, 3.40 \cdot 10^6$	$3.80, 5.54 \cdot 10^4$	108, 61	$1.40 \cdot 10^9$	$1.63 \cdot 10^6$	31,225 / 45,343
197	$4.91, 2.83 \cdot 10^6$	$1.40 \cdot 10^5, 5.44 \cdot 10^4$	35, 52	$1.41 \cdot 10^9$	$1.65 \cdot 10^6$	142,361 / 74,345
269	$5.51, 2.57 \cdot 10^6$	$2.30, 2.34 \cdot 10^4$	240, 110	$1.41 \cdot 10^9$	$1.65 \cdot 10^6$	44,281 / 50,040

397	—	$1.31, 1.58 \cdot 10^4$	—	—	$1.66 \cdot 10^6$	66,981 / 89,424
521	—	$1.88, 2.15 \cdot 10^4$	—	—	$1.66 \cdot 10^6$	131,952 / 142,487

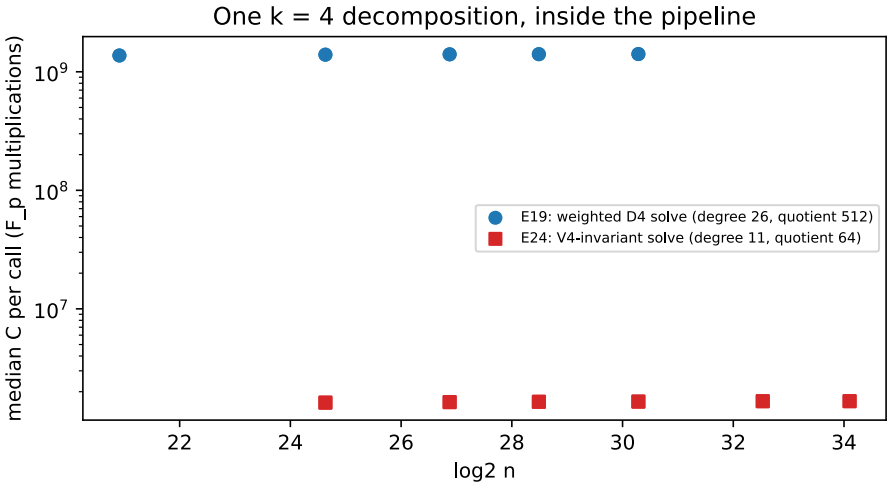


Figure 2: Median C per call inside the pipelines: weighted D_4 (quotient 512) against the V_4-invariant solve (quotient 64).

3. Predictions and verdicts

id	registered	measured
G2.9–G2.11	E19 correct; $y \in [70, 160]$; C within ±30 % of d4-s5	hold: all logs; $y = 115.1$; $C = 1.37$–$1.41 \cdot 10^9$
G2.12	E19 $S/$ rho within ×2 of $0.584C/p$; slope $[-0.35, -0.15]$	per size holds (one curve at 0.495); slope -0.13: falsified
G2.13	E19 handover within ±3 bits of $2^{118.9}$	holds: $2^{116.7}$
G2.14–G2.17	E20: DLP dearer at measured sizes; crossovers $> 2^{160}$	hold: 1.7–3.6 × dearer; 2^{164}–2^{222}
G2.18–G2.21	E21: replay 1.03–1.20 ×, $\leq 5\%$ fallbacks	agreement holds; gain and fallbacks falsified (net loss)
G2.22–G2.24	E22: line exists; C down 4–30 ×; y up 3–10 ×	line holds; C down ≈ 820 × (falsified high); y up ≈ 13 ×
G2.26	E23: no useful shared base on $j = 0$	holds (confined or empty)
G2.27–G2.31	E24: correct; $C \in [1.3, 2.2] \cdot 10^6$; 20–300 × below E19; slope; $n^* \in [2^{86}, 2^{98}]$	all hold: 35–240 ×; slope -0.25; $n^* = 2^{90.1}$

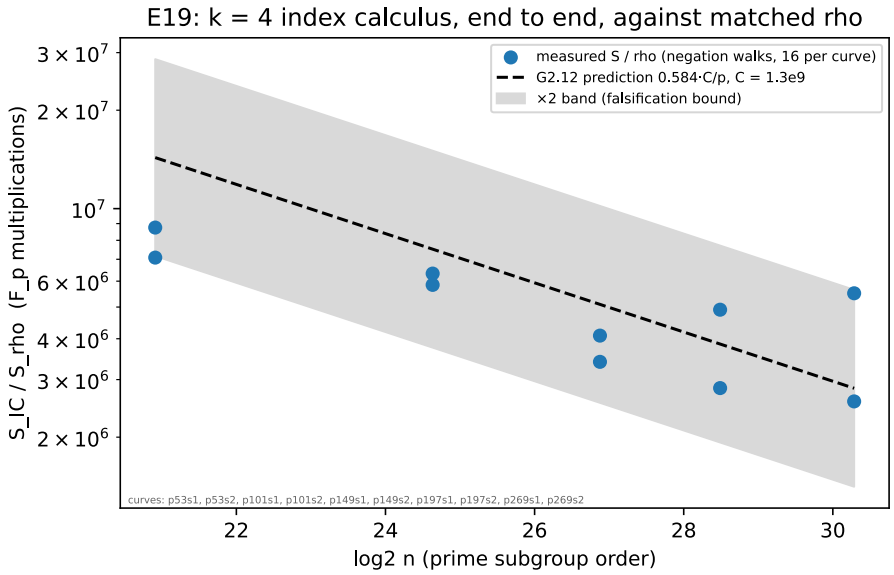


Figure 3: E19 against G2.12's registered prediction and its ×2 band (10 curves).

4. Records

Raw records and tables: `experiments/27_quartic_pipeline_d4/` (E19), `experiments/28_e20_dlp/`, `experiments/28_e21_replay/`, `experiments/28_e22_stage_a/`, `experiments/28_e22_stage_b/`, `experiments/29_quartic_pipeline_v4/` (E24). Code: `src/cryptanalysis/quartic_folds.rs`, `src/cryptanalysis/klein_v4.rs`, `src/cryptanalysis/d4_weighted_s5.rs` (trace, additive); runners `examples/quartic_pipeline_{d4,v4}.rs`, `examples/klein_stage_{a,b}.rs`, `examples/d4_replay.rs`, the ladder's `--arms d4dlp`. Tables: `examples/glv_invariant_experiment_tables.rs`.

5. Errata (2026-10-21)

An independent validator replayed every E19 and E24 run (field-identical apart from wall time, every logarithm confirmed) and recomputed the numbers above from the raw JSON (`benchmarks/20261018-k4-v4-line/validator-report.json`). The text above is left as written; these values supersede it. No verdict changes. Full list: the note's §9.19.

where	written	raw data
E19 / E24, $p = 101$ seed 2	171 ×	170 × (170.4; 171 came from dividing rounded values)
G2.12, $p = 53$ seed 1	0.495	0.494 (0.49448)

At 0 GAE per unpriced native operation the E19-to-E24 factor is 35–240 ×. Pricing one GAE (97 F_p multiplications) per Macaulay-matrix entry and per walk step, it is 19–132 × at 0.1 GAE and 6.5–45 × at 1 GAE; E24 stays below E19 on every curve at every price.